



In silico identification and expression analysis of Rare Cold Inducible 2 (RCI2) gene family in cucumber

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Received: 12 September 2018 / Accepted: 17 May 2019 / Published online: 5 June 2019
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Abstract

Rare Cold Inducible 2 (RCI2), also known as plasma membrane protein 3, is a group of low molecular weight proteins that play crucial roles in plant growth and development as well as environmental stress responses. In this work, we performed a comprehensive analysis of the *RCI2* gene family and identified 4 *RCI2* genes (*CsaRCI2A–D*) from cucumber (*Cucumis sativus* L.), which were distributed in four chromosomes of the cucumber genome. The amino acid identities among the 4 *CsaRCI2* proteins varied from 46.55 to 76.00%. Phylogenetic analysis showed that *RCI2* proteins from cucumber, *Arabidopsis*, *Medicago truncatula*, maize and rice could be classified into four distinct clades, and nearly all the *RCI2* genes harbored two exons separated by an intron of variable size. Many development-, stress-, and hormone-related *cis*-elements were found in the promoter regions of the *CsaRCI2* genes, and the *CsaRCI2* genes seemed to have tissue-specific expression patterns. In addition, qRT-PCR analysis revealed that these four *CsaRCI2* genes were all responsive to various abiotic stresses including cold, salt, and drought, as well as ABA treatment. Our findings lay a foundation for further elucidating the biological roles of the *RCI2* gene family in cucumber.

Keywords Cucumber · *RCI2* · PMP3 · Phylogenetic analysis · Expression pattern · Abiotic stress

Abbreviations

RCI2	Rare Cold Inducible 2
Pmp3	Plasma membrane protein 3
Lti	Low temperature induced
MW	Molecular weight
CDS	Coding sequence
gDNA	Genomic DNA
qRT-PCR	Quantitative real-time PCR

Introduction

Different abiotic stresses, such as high salinity, drought, and extreme temperature, can adversely affect plant growth and development. Exposure to these stresses can disrupt ion homeostasis by impairing the membrane permeability and inducing electrolyte leakage through the cell (Yeshvekar et al. 2017). Many integral membrane proteins have been suggested to facilitate transport of ions and other small molecules across the membrane, and maintain the membrane potential and re-establish ion homeostasis under various abiotic stresses. Among these proteins, Rare Cold Inducible 2 (RCI2) proteins were shown to be active in cellular ion homeostasis and function in the maintenance of plasma membrane potential (Navarre and Goffeau 2000).

RCI2 genes, which are often named as *plasma membrane proteins 3* (PMP3) or *low temperature induced 6* (*Lti6*) genes, are present as multigene families in plants. Most of *RCI2* genes have two exons and one intron, and encode small molecular weight (MW) and highly hydrophobic proteins with the PMP3 domain (PF01679) (Rocha 2016; Medina et al. 2001; Morsy et al. 2005). Plant

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s13562-019-00510-6>) contains supplementary material, which is available to authorized users.

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RCI2 proteins possess two putative α -helix domains and can be classified into two groups based on their C-terminal tails. Group I RCI2s are shorter proteins (about 52–64 aa), while Group II RCI2s (approximately 60–89 aa) are longer than Group I RCI2s with hydrophilic C-terminal extensions and are rich in charged residues (Medina et al. 2007). In addition, Group I RCI2s can be further divided into two sub-groups (Ia and Ib) on the basis of their C-terminal hydropathicity: Group Ia RCI2s possess hydrophobic C-terminal ends while Group Ib RCI2s contain hydrophilic C-terminal ends (Rocha 2016; Yeshvekar et al. 2017).

The first RCI2 protein (PMP3) was functionally identified in yeast (*Saccharomyces cerevisiae*) and the *pmp3* mutant yeast cells exhibited phenotypes of plasma membrane hyperpolarization, increased Na^+ influx, and salt sensitivity (Navarre and Goffeau 2000). Many RCI2 homologs can complement the *PMP3* deletion in yeast under salinity stress, such as AtRCI2A, AtRCI2B, AtRCI2C and AtRCI2H from *Arabidopsis* (Nylander et al. 2001; Medina et al. 2007), AcPMP3-1 from *Aneurolepidium chinense* (Inada et al. 2005), OsLTI6a and OsLTI6b from rice (Zhang et al. 2008), PutPMP3-1 and PutPMP3-2 from *Puccinellia tenuiflora* (Zhang et al. 2008), ZmPMP3s (except for ZmPMP3-6) from maize (Fu et al. 2012), MpRCI from *Musa paradisica* (Liu et al. 2012), MsRCI2A from *Medicago sativa* (Long et al. 2015), MtRCI2A, MtRCI2B and MtRCI2C from *Medicago truncatula* (Long et al. 2015), and CsRCI2A from *Camelina sativa* (Kim et al. 2016), suggesting that these homologs share similar functions in maintaining membrane potential and regulating ion homeostasis under salt stress. For example, overexpression of *AtRCI2A* in *Arabidopsis* plants resulted in restricted Na^+ uptake into cells and therefore mitigated damage from salt stress (Mitsuya et al. 2006), while antisense repression of *AtRCI2A* led to an excessive accumulation of Na^+ and increased salt sensitivity (Mitsuya et al. 2005). Overexpression of *ZmPMP3.1* in *Arabidopsis* enhanced plant growth under salt stress by affecting the expression of some ion transporter genes and possibly regulating membrane potential (Fu et al. 2012). Overexpression of *MpRCI* in *AtRCI2A* knockout mutant also resulted in an increase in NaCl tolerance by protecting plasma membrane (PM) fluidity and H^+ -ATPase activity (Liu et al. 2012). Besides salt stress, transgenic plants overexpressing some *RCI2* genes demonstrated enhanced tolerance to other abiotic stresses, including heat, cold, and drought (Ben Romdhane et al. 2017; Khurana et al. 2015; Li et al. 2014; Kim et al. 2007; Ben-Romdhane et al. 2018). As a whole, these studies confirm that *RCI2* genes play important roles in abiotic stress responses.

As an important vegetable crop grown worldwide, cucumber is susceptible to a number of abiotic stresses, which cause severe losses in its yield (Zhou et al. 2018b;

Bai et al. 2018). Although *RCI2* genes play important roles in plant stress responses, no comprehensive study has been conducted on the cucumber *RCI2* gene family so far. In this study, we identified 4 *RCI2* family genes in cucumber, and analyzed their chromosomal distributions, gene structures, phylogenetic relationships, *cis*-elements in promoter regions, and tissue expression patterns. Additionally, we also investigated the expression patterns of these genes under exposure to cold, salt, drought and ABA treatments. The findings will facilitate the functional analysis of *RCI2* genes in cucumber and thus provide theoretical basis for further genetic improvement of cucumber in future studies.

Materials and methods

Identification and sequence analysis of RCI2 members in cucumber

Arabidopsis and *M. truncatula* RCI2 protein sequences were extracted from the Arabidopsis Information Resource (TAIR, <http://www.arabidopsis.org/>) and the *M. truncatula* genome database (<https://phytozome.jgi.doe.gov/Mtruncatula>), respectively, based on the protein IDs in previous studies (Medina et al. 2007; Long et al. 2015). Then, these sequences were used as queries in the BLASTP search of the cucumber genome database (<http://cucurbitgenomics.org/organism/2>) with an e value cutoff of $1e^{-5}$. Additionally, the Hidden Markov Model (HMM) profile of the PMP3 domain (PF01679) was downloaded from the Pfam database (<http://pfam.xfam.org/>) and used for searching the cucumber genome database using HMMER3.0 (Eddy 2011). Each non-redundant sequence was checked for the presence of the PMP3 domain by using NCBI CDD (<https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>) and SMART (<http://smart.embl-heidelberg.de/>).

The physicochemical features including isoelectric point (pI), molecular weight (MW) and grand average of hydropathy index (GRAVY) for each CsaRCI2 protein were investigated by using the ProtParam server (<http://web.expasy.org/protparam/>). ProtComp 9.0 (<http://www.softberry.com/berry.phtml>) was employed for predicting the subcellular localization of each CsaRCI2 protein. The MEME server (<http://meme-suite.org/tools/meme>) was used to examine the amino acid positions of conserved domains of CsaRCI2 proteins. For transmembrane (TM) domain analysis, the amino acid sequences of CsaRCI2 proteins were examined by the TMHMM v.2.0 (<http://www.cbs.dtu.dk/services/TMHMM/>) and SOSUI (http://harrier.nagahama-i-bio.ac.jp/sosui/sosui_submit.html). Chromosomal positions (start and end position) of the *CsaRCI2* genes were retrieved from the cucumber genome database, and the MapInspect software was employed to draw the their

positions on chromosomes as described previously (Zhou et al. 2018a). The segmental and tandem duplication genes were defined based on the criteria as previously described (Kong et al. 2013; Kayum et al. 2017).

Multiple sequence alignments and phylogenetic tree construction

The amino acid sequences of RCI2 proteins from cucumber, *M. truncatula*, and *Arabidopsis* were aligned using Clustal Omega with default settings (Sievers et al. 2011), and then visualized and redrawn by using GeneDoc. We also downloaded the RCI2 protein sequences of rice and maize based on the accession numbers in previous studies (Medina et al. 2007; Zhao et al. 2014) from the rice genome annotation project database (<http://rice.plantbiology.msu.edu/>) and the maize genome database (<https://phytozome.jgi.doe.gov/maize>), respectively. Then, multiple sequence alignments were performed using Clustal Omega with the default parameters for all of the 4 *CsaRCI2s*, together with 8 *AtRCI2s*, 4 *MtRCI2s*, 9 *OsRCI2s*, and 10 *ZmRCI2s*. Based on the alignment results, a phylogenetic tree was created using the MEGA 5.0 software by neighbor-joining (NJ) approach (Tamura et al. 2011), and bootstrap analysis was carried out with 1000 replicates. The proteins and their accession numbers used for alignment and phylogenetic tree generation are presented in Table S1.

Gene structure and cis-element analysis of *CsaRCI2* genes

The coding sequence (CDS) and corresponding genomic DNA (gDNA) of *RCI2* genes of cucumber, *M. truncatula*, *Arabidopsis*, maize, and rice were retrieved from the cucumber genome database (<http://cucurbitgenomics.org/organism/2>), the *M. truncatula* genome database (<https://phytozome.jgi.doe.gov/Mtruncatula>), the *Arabidopsis* Information Resource (TAIR, <http://www.arabidopsis.org/>), the maize genome database (<https://phytozome.jgi.doe.gov/maize>), and the rice genome annotation project database (<http://rice.plantbiology.msu.edu/>), respectively. Subsequently, the GSDS online tool (<http://gsds.cbi.pku.edu.cn>) was used to examine the gene structure of each *RCI2* gene. For cis-elements analysis, 1000-bp upstream sequences of the start codons of *CsaRCI2* genes obtained from the cucumber genome database were identified as putative promoter regions, and analyzed using online software PlantCARE with default settings (Lescot et al. 2002).

Tissue expression analysis of *CsaRCI2* genes

The digital gene expression in different cucumber tissues, including female flower, leaf, male flower, expanded ovary with fertilization, unexpanded ovary, expanded ovary without fertilization, root, stem, base part of tendril, and tendril, was downloaded based on the transcriptome data as previously reported (Li et al. 2011), which is available in the NCBI Sequence Read Archive (SRA) database (<https://www.ncbi.nlm.nih.gov/sra/?term=sra046916>). The FPKM (Fragments Per Kilobase of exon model per million mapped reads) values of *CsaRCI2* genes were picked out and the heat maps were generated as previously described (Zhou et al. 2018a).

Plant materials and growth conditions

Plants of cucumber (*Cucumis sativus* L. cv ‘Chinese long’ 9930 inbred line) were planted in trays containing soil mixture (peat:sand:pumice, 1:1:1, v/v/v) at 24/18 °C (day/night) under a 16 h light/8 h dark photoperiod. Two-week-old cucumber seedlings were treated with 200 mM NaCl and 4 °C to induce salt stress and cold stress, respectively, according to the description in our previous study (Zhou et al. 2018a). For drought and abscisic acid (ABA) treatment, 2-week-old seedlings were grown in liquid Murashige and Skoog (MS) medium containing 10% PEG-6000 (w/v) and 100 µM ABA, respectively. The leaf samples were then harvested at 0, 3, 6 and 12 h after abiotic treatments, and stored at − 80 °C until RNA extraction.

RNA extraction, cDNA synthesis and quantitative real-time PCR (qRT-PCR)

Total RNA was extracted using RNA prep Pure Plant Kit (TransGen, China) following the manufacturer’s instructions. The TransScript First-Strand cDNA Synthesis SuperMix Kit (TransGen, China) was used to synthesize first-strand cDNA from the extracted RNA digested by DNase I according to the manufacturer’s instruction. Quantitative real-time PCR (qRT-PCR) assays were performed with the SYBR Premix ExTaq kit (TaKaRa, Japan) on the LightCycler 480 real time system. The reaction conditions were set as follows: 95 °C for 5 min, followed by 40 cycles of 95 °C for 10 s and 60 °C for 30 s. Each reaction of the qRT-PCR was performed by triplicate, and the relative expression levels were computed using $2^{-\Delta\Delta Ct}$ method (Livak and Schmittgen 2001), by using *CsAct3* (GenBank accession no. DQ115883) as internal reference. The primers used for qRT-PCR analysis are listed in Table S2.

Results

Identification of the *RCI2* gene family in cucumber

A total of 4 *RCI2* genes were identified from the whole genome of cucumber, which were named as *CsaRCI2A* to *CsaRCI2D* based on their order of distribution on the chromosomes (Table 1). The gDNA length of *CsaRCI2* genes varied from 307 bp (*CsaRCI2D*) to 1811 bp (*CsaRCI2C*), which encoded peptides ranging from 54 to 78 aa with the pI values from 4.25 to 8.14, and the MW from 5.95 to 8.69 kDa (Table 1). In addition, the GRAVY values of *CsaRCI2* proteins ranged from 0.897 (*CsaRCI2A*) to 1.620 (*CsaRCI2D*), suggesting that they are highly hydrophobic. Moreover, the prediction of subcellular localization by ProtComp 9.0 indicated that *CsaRCI2A*, *CsaRCI2C* and *CsaRCI2D* were located in the plasma membrane, while *CsaRCI2B* was located in the vacuole (Table 1).

In silico mapping of the gene loci showed that the *CsaRCI2* genes were located in four different chromosomes in cucumber genome, with one gene on each of chromosomes 1, 2, 3 and 5 (Fig. 1). No segmental or tandem duplication events were determined based on the criteria of previous studies (Kong et al. 2013; Kayum et al. 2017).

Characterization of deduced *CsaRCI2* proteins

Alignment analysis of amino acids showed that the identities between *CsaRCI2* proteins ranged from 46.55% (*CsaRCI2B* and *CsaRCI2C*) to 76.00% (*CsaRCI2A* and *CsaRCI2B*). To study the features of the homologous domain sequences, multiple sequence alignments were carried out based on the *RCI2* protein sequences of cucumber, *M. truncatula*, *Arabidopsis*, maize and rice. The results showed that the *RCI2* proteins comprised two conserved hydrophobic domains separated by a putative

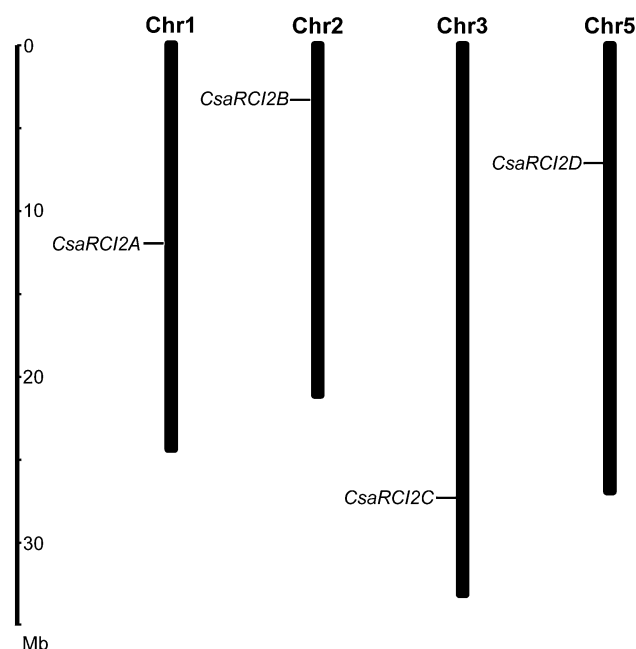


Fig. 1 Chromosomal locations of *CsaRCI2* genes. Gene positions and chromosome sizes are shown in Megabase (Mb)

loop (Fig. 2). In addition, *CsaRCI2A* and *CsaRCI2B* possessed longer hydrophilic C-terminal tails (about 20 amino acids) with charged residues, suggesting that they belong to the Group II *RCI2* proteins. An extra conserved Cys residue was also found between the two conserved hydrophobic domains in the majority of Group II *RCI2* proteins (Fig. 2). Other *RCI2*s were shorter proteins with a small C-terminal tail, indicating that they belong to the Group I *RCI2* proteins. Amongst them, *CsaRCI2C* showed a high identity to Group Ib *RCI2* proteins including *AtRCI2A-C* and *MtRCI2A-C*, while a high similarity was observed between *CsaRCI2D* and *AtRCI2H*, indicating that *CsaRCI2C* and *CsaRCI2D* are Group Ib and Group Ia *RCI2* proteins, respectively.

MEME was employed to predict the conserved motifs in *CsaRCI2* proteins, and a total of three conserved motifs

Table 1 Basic characteristics of the *RCI2* gene family in cucumber

Gene	Gene ID	Genomic position	gDNA (bp)	CDS (bp)	Length (aa)	PI	MW (kDa)	GRAVY	Subcellular prediction
<i>CsaRCI2A</i>	Csa1G257300	Chr1: 12585784..12586633 (+)	850	228	75	5.09	8.61	0.897	Plasma membrane
<i>CsaRCI2B</i>	Csa2G036040	Chr2: 3672581..3673361 (+)	781	237	78	4.33	8.69	1.047	Vacuole
<i>CsaRCI2C</i>	Csa3G736860	Chr3: 28454898..28456708 (–)	1811	210	69	8.14	7.67	1.054	Plasma membrane
<i>CsaRCI2D</i>	Csa5G175890	Chr5: 7469237..7469543 (+)	307	165	54	4.25	5.95	1.620	Plasma membrane

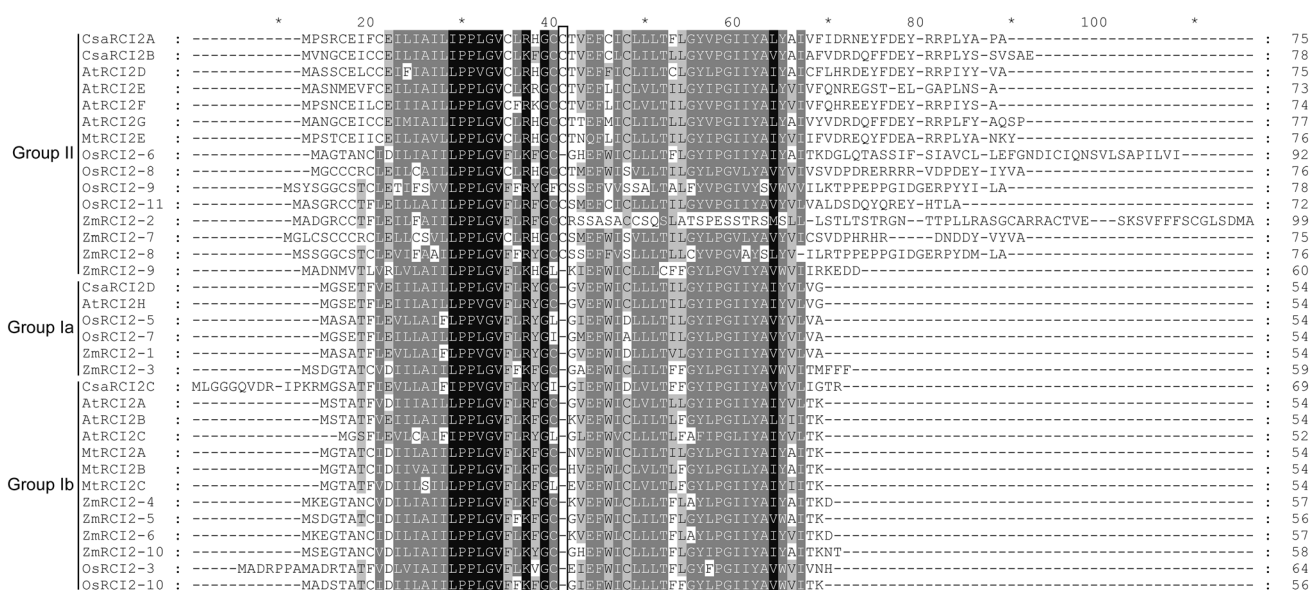


Fig. 2 Alignment of RCI2 protein sequences from cucumber, *M. truncatula*, *Arabidopsis*, maize and rice. Multiple sequence alignment was conducted using Clustal Omega and visualized in GeneDoc. Two

conserved hydrophobic regions predicted to be transmembrane domains with TMHMM 2.0 and SOSUI were underlined. The extra conserved Cys residue in the Group II RCI2s was boxed

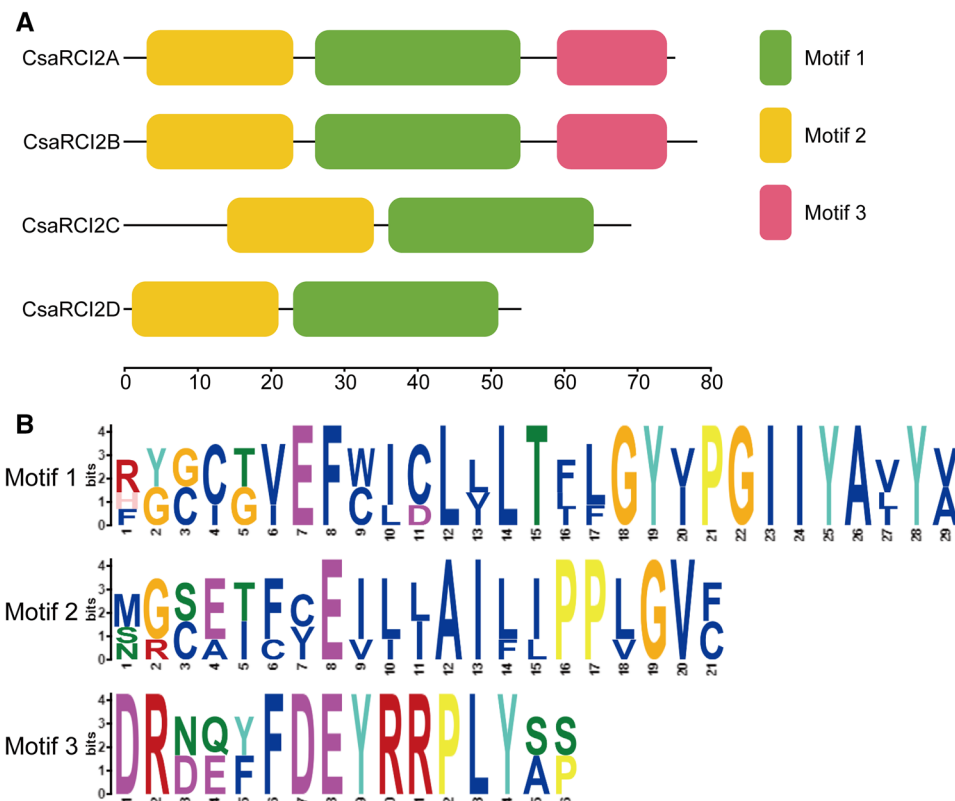
were identified. Motifs 1 and 2 are related to the PMP3 domain, which are present in all CsaRCI2 proteins (Fig. 3). In addition, motif 3 is present in Group II CsaRCI2 proteins (CsaRCI2A and CsaRCI2B) but absent in Group I

CsaRCI2 proteins (CsaRCI2C and CsaRCI2D) (Fig. 3), which further supported the group classification (Fig. 2).

Fig. 3 Conserved motif analysis of CsaRCI2 proteins.

a Distribution of conserved motifs among CsaRCI2 proteins analyzed by MEME.

b Distribution of amino acid positions of conserved motifs for CsaRCI2 proteins



Phylogenetic and gene structure analysis of *RCI2* genes in cucumber and other plant species

To elucidate the evolutionary relationship of *RCI2* genes in cucumber and other plant species, a phylogenetic tree was constructed using the protein sequences of 4 *CsaRCI2* genes, 8 *AtRCI2* genes, 4 *MtRCI2* genes, 10 *ZmRCI2* genes, and 9 *OsRCI2* genes. In this analysis, the *RCI2* proteins were categorized into four distinct clades, and Clade a contained the most members (14), while Clade d contained the fewest (6) (Fig. 4a). Interestingly, *CsaRCI2* proteins were only found in Clades b and c. Amongst them, *CsaRCI2A* was homologous to *MtRCI2E* and *AtRCI2F*, *CsaRCI2B* homologous to *AtRCI2G*, *CsaRCI2C* homologous to *ZmRCI2-1* and *OsRCI2-5*, and *CsaRCI2D* homologous to *AtRCI2H* (Fig. 4a). In addition, all the members in Clade b and d were exclusively from dicots and monocots, respectively, while Clade a and c included *RCI2* proteins from both dicots and monocots (Fig. 4a), indicating that the separation of *RCI2* genes might have occurred after the divergence of monocots and dicots.

We additionally performed a gene structure analysis by aligning the CDS sequences with the corresponding gDNA sequences of cucumber, *M. truncatula*, *Arabidopsis*, maize and rice. Based on their gene structures, nearly all *RCI2* genes exhibited a structure of two exons and one intron, with the exception of *OsRCI2-6* and *OsRCI2-1*, which contained two and four introns, respectively (Fig. 4b). The lengths of untranslated regions (UTRs) varied among these genes, but six members (*OsRCI2-6*, *MtRCI2B*, *MtRCI2C*, *CsaRCI2D*, *OsRCI2-9*, and *OsRCI2-1*) were predicted to have no UTRs (Fig. 4b).

Promoter region analysis of *CsaRCI2* genes

The identification of *cis*-acting regulatory elements in the promoter region can provide insights into the tissue-specific or stress-responsive expression patterns of genes (Liu et al. 2017; Bitas et al. 2016). In order to investigate the potential expression profile of *CsaRCI2* genes, we analyzed the 1.0-kb genomic sequences upstream their start codons by using online PlantCARE software. A number of development-, stress-, and hormone-related *cis*-elements

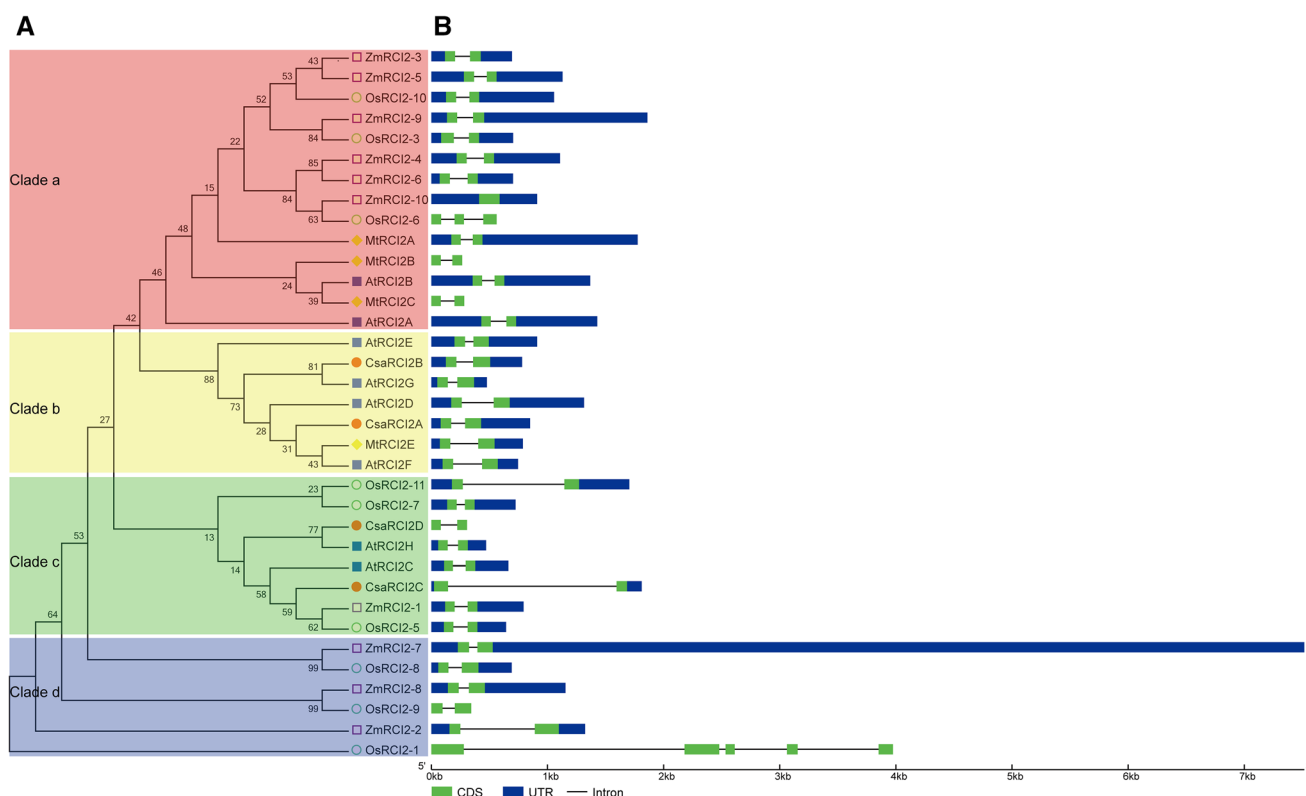


Fig. 4 Phylogenetic relationships and gene structures of *RCI2* genes in cucumber, *M. truncatula*, *Arabidopsis*, maize and rice. **a** Phylogenetic relationships of the *RCI2* proteins in cucumber, *M. truncatula*, *Arabidopsis*, maize and rice. The phylogenetic tree was constructed using MEGA 5.0 software by the NJ method and bootstrapped 1000 times. Bootstrap values are shown at each node. **b** Gene structures of

the *RCI2* genes in cucumber, *M. truncatula*, *Arabidopsis*, maize and rice were determined by aligning the CDS sequences with the corresponding gDNA sequences using the GSDS Server. The green bars, blue bars and plain lines indicate CDSs, UTRs (untranslated regions), and introns, respectively

were identified in the promoter regions of the *CsaRCI2* genes (Fig. 5; Table S3). Among these *cis*-elements, hormone-related *cis*-elements were the most abundant, such as CGTCA-motif, AuxRR-core, TGA-element, ABRE, ERE, GARE-motif, and TCA-element, indicating their roles in response to different hormones, including MeJA, auxin, ABA, ethylene, GA, and SA (Narusaka et al. 2003; Anderson et al. 1994; Nejad et al. 2012; Shen and Ho 1995; Goldsbrough et al. 1993). In addition, all these genes had at least three types of hormone-related *cis*-elements in their promoter regions. Six *cis*-elements were found to be stress-responsive elements, including HSE, MBS, TC-rich repeats, W-box, WUN-motif and ARE, which are involved in heat stress responsiveness, drought inducibility, defense and stress responsiveness, abiotic stress and defense responsiveness, wound responsiveness, and anaerobic induction, respectively. Moreover, the *CsaRCI2* promoters also contained four kinds of development-related *cis*-elements, including CAT-box, O2-site, CCGTCC-box, and Skn-1_motif (Fig. 5; Table S3), which are related to meristem expression, zein metabolism regulation, meristem specific activation, and endosperm expression, respectively. These results suggested that the *CsaRCI2* genes might be involved in tissue development, as well as in responses to multiple hormones and stresses.

Tissue-specific expression of *CsaRCI2* genes in cucumber

The expression levels of *CsaRCI2* genes in different tissues were examined using online transcriptome data. Surprisingly, the expression of *CsaRCI2D* was undetectable in all the tested tissues, indicating that it may have some expression patterns (Fig. 6). *CsaRCI2A* was highly expressed in male and female flowers, moderately expressed in leaves, tendrils and tendril bases, and weakly expressed in other tissues. *CsaRCI2B* had the highest transcription in tendril bases, and relatively lower transcription in male and female flowers. *CsaRCI2C* showed

higher expression in fertilized ovaries and significantly lower expression in male flowers, unexpanded ovaries and female flowers, and was not expressed in other tissues (Fig. 6).

Expression patterns of *CsaRCI2* genes under abiotic stress and ABA treatment

The expression levels of *CsaRCI2* genes under different abiotic stresses (cold, salt, and drought) and ABA treatment were assessed by qRT-PCR. As shown in Fig. 7a, the expression levels of all *CsaRCI2* genes were up-regulated at the beginning of cold stress, and finally decreased at 12 h after treatment. Under salt treatment, *CsaRCI2A*, *CsaRCI2B*, and *CsaRCI2C* exhibited up-regulated expression at all the time points, while the expression of *CsaRCI2D* was only increased at 12 h after treatment (Fig. 7b). After drought treatment, all *CsaRCI2* genes displayed the same trend of up-regulation at all time points, with a remarkable increase at 3 h and 6 h, followed by a decrease at 12 h (Fig. 7c). Upon ABA treatment, the expression of all *CsaRCI2* genes was significantly up-regulated at 3 h, but obviously down-regulated at certain time points after treatment (Fig. 7d).

Discussion

Genome-wide identification and characterization of *RCI2* genes have been carried out in various plant species, such as *Arabidopsis thaliana* and *Oryza sativa* (Medina et al. 2007), *Zea mays* (Zhao et al. 2014; Fu et al. 2012), and *M. truncatula* (Long et al. 2015), but not in cucumber, which has a high transpiration rate and is susceptible to various abiotic stresses. In this study, a total of 4 *RCI2* genes were identified in cucumber genome, and no segmental or tandem duplication events were found between the *CsaRCI2* genes (Table 1; Fig. 1), indicating a relatively slow evolutionary process of the *RCI2* gene family in cucumber.

Fig. 5 *Cis*-acting regulatory elements in the promoter regions of *CsaRCI2* genes. The number of *cis*-elements in the promoter region (1 kb upstream of the translation start site) of each *CsaRCI* gene was exhibited by different colors and numbers in the grid

	Development				Stress						Hormone						
	CAT-box	O2-site	CCGTCC-box	Skn-1_motif	HSE	MBS	TC-rich repeats	W-box	WUN-motif	ARE	CGTCA-motif	AuxRR-core	TGA-element	ABRE	ERE	GARE-motif	TCA-element
CsaRCI2A				1						2		1	2	2			1
CsaRCI2B				1					1	2			1		1	2	
CsaRCI2C	2	1		1	1	3	1	1		1	2					1	1
CsaRCI2D			1	1		2	1			3				3		1	2

Fig. 6 Expression profiles of *CsaRCI2* genes in different cucumber tissues by transcriptome data. The gene expression levels are represented by different colors from blue to red

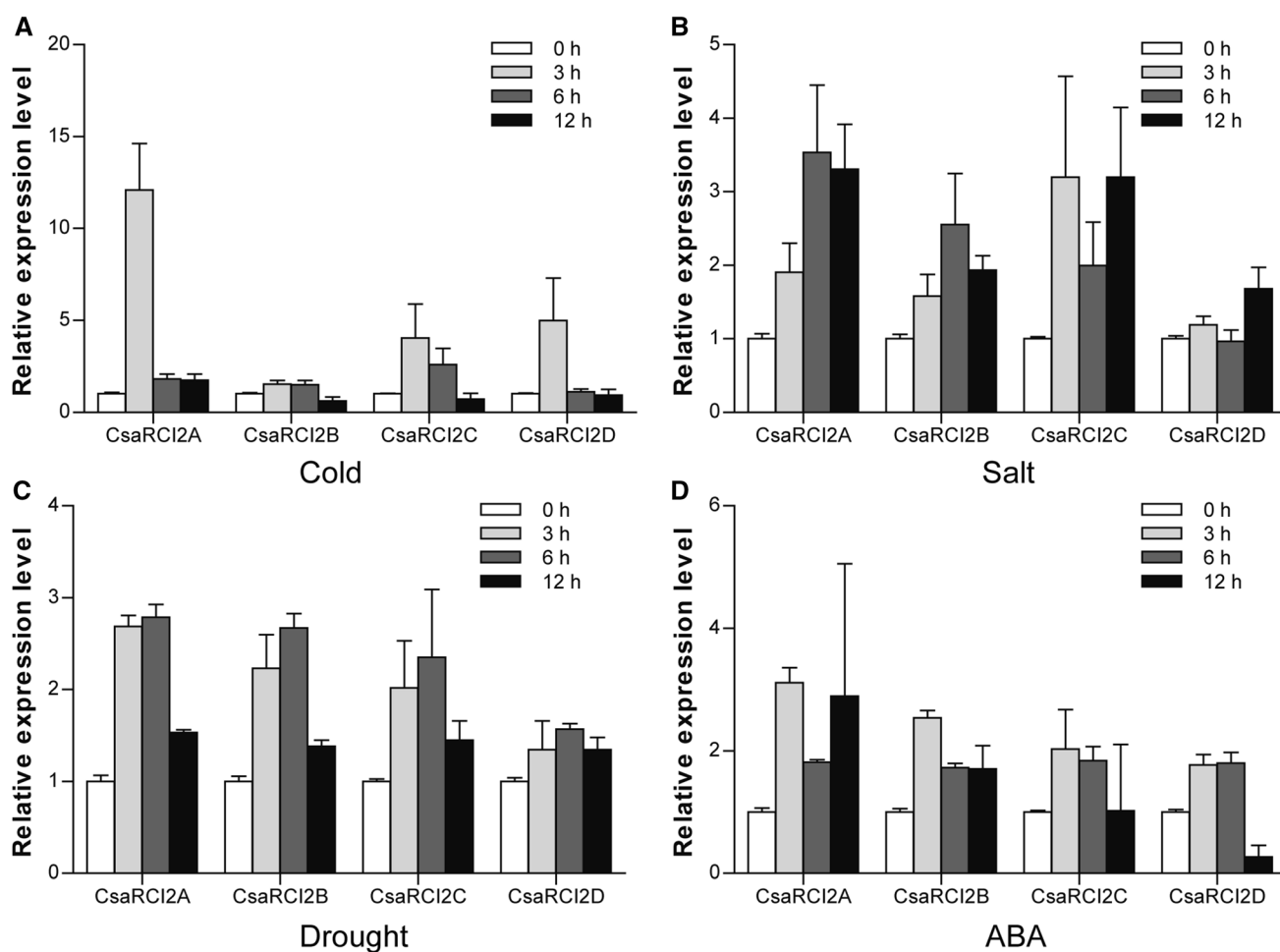
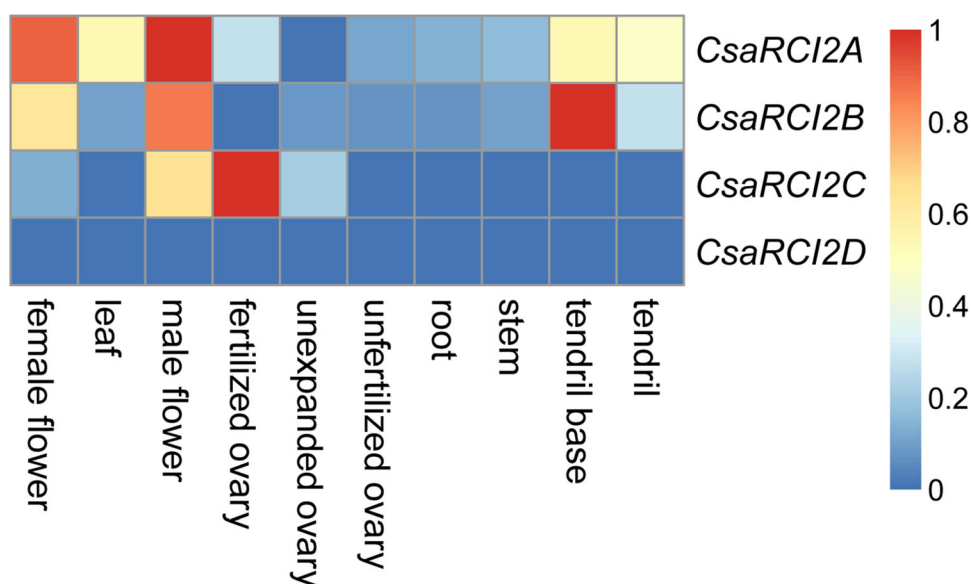


Fig. 7 Expression profiles of *CsaRCI2* genes in response to various stress conditions including cold (a), salt (b), drought (c), and ABA (d). The relative expression levels of *CsaRCI2* genes were analyzed at 3, 6, and 12 h of treatments as compared with their values at 0 h

RCI2 proteins can function in distinct membranes besides plasma membrane (PM). For example, some RCI2 proteins were found to be located in the endoplasmic reticulum, Golgi apparatus, intracellular membranes and punctate structures (Pokrzywa et al. 2009; Medina et al. 2007; Long et al. 2015). In this study, *CsaRCI2A*, *CsaRCI2C* and *CsaRCI2D* were predicted to be located in the plasma membrane, while *CsaRCI2B* was located to the vacuole (Table 1). Considering that the vacuole is the main intracellular compartment for ion homeostasis and sequestration of Na^+ (Blumwald 2000), *CsaRCI2B* may have a particular function in cucumber. In addition, sequence analysis indicated that *CsaRCI2C* and *CsaRCI2D* belong to Group Ib and Group Ia RCI2 proteins, respectively. However, *CsaRCI2C* displayed longer N- and C-terminal than other Group I RCI2 proteins (Fig. 2), implying the existence of splice variants of the *CsaRCI2C* gene in cucumber. Similar splicing events were identified in *Nicotiana tabacum* (Kumari et al. 2017).

Previous studies have shown that many plant RCI2 genes have organ- and tissue-specific expression patterns (Rocha 2016; Chidambaram et al. 2015), indicating that these genes play essential roles in specific tissues. For example, *AtRCI2H* displayed a noteworthy expression-specificity in seeds (Medina et al. 2007). Similarly, the expression of *ZmRCI2-1* is restricted in germinating seeds (Zhao et al. 2014). In the present study, nearly all the *CsaRCI2* genes showed the highest expression levels in some specific tissues including flowers, tendril bases, and fertilized ovaries (Fig. 6), implying the important roles of *CsaRCI2* genes in the development of these tissues.

Plasma membrane is the first barrier when a plant is exposed to abiotic stresses, and accumulating evidence indicates that RCI2 genes are induced by abiotic stresses, particularly salt and cold stress (Chidambaram et al. 2015; Rocha 2016; Kim et al. 2016). For instance, the expression of *AcPMP3-1* and *AcPMP3-2* was highly induced under a series of abiotic stresses, such as salt, cold and drought (Inada et al. 2005). The vast majority of maize RCI2 genes are also responsive to salt and drought stresses with differential expression patterns (Zhao et al. 2014; Fu et al. 2012). In this study, all the *CsaRCI2* genes were found to be responsive to various abiotic stresses to different degrees (Fig. 7), and a series of stress-related *cis*-elements were found in the promoter regions of *CsaRCI2* genes (Fig. 5; Table S3), indicating their roles in response to multiple abiotic stresses. Similar results were obtained in *P. tenuiflora* (*PutPMP3-1* and *PutPMP3-2*) (Zhang et al. 2008), *Salvia miltiorrhiza* (*SmPMP3-1* and *SmPMP3-2*) (Wang et al. 2013), and *Pennisetum glaucum* (*PgPMP3-1* and *PgPMP3-2*) (Yeshvekar et al. 2017). In addition, *CsaRCI2* genes were induced very early under stresses, especially under cold treatment (Fig. 7a), implying their important

roles in response to cold stress. Under cold treatment, *OsLti6a* was induced very early in a cold-tolerant rice line, but the induction was delayed by 12 h in a cold-sensitive rice line (Morsy et al. 2005). Surprisingly, *CsaRCI2A*, *CsaRCI2C*, and *CsaRCI2D* were up-regulated at 3 h under cold treatment, but no previously identified cold-responsive *cis*-elements were found for these genes (Figs. 5, 7a), implying that there may be some unknown cold-responsive *cis*-elements in their promoter regions. Moreover, different *CsaRCI2* genes exhibited various expression profiles under stress treatments, implying that they may function in response to different stresses. For example, the highest relative expression level of *CsaRCI2A* and *CsaRCI2D* was detected after cold stress treatment, whereas the highest expression of *CsaRCI2B* and *CsaRCI2C* was observed after salt stress treatment (Fig. 7a, b). Recent studies have revealed that the expression patterns of *AITMP1* and *AITMP2* are different under salt and osmotic stresses, and overexpression of *AITMP2* in tobacco resulted in higher tolerance to salt stress than to drought stress, while it was just the opposite case for the overexpression of *AITMP1* (Ben Romdhane et al. 2017; Ben-Romdhane et al. 2018). Both *AITMP1* and *AITMP2* confer tolerance to a variety of stresses in tobacco through maintaining ion homeostasis and increasing membrane integrity, suggesting that RCI2 genes are involved in the maintenance of plasma membrane potential under stress. In a previous study, overexpression of *ZmPMP3.1* in *Arabidopsis* conferred salt stress tolerance by affecting the expression of some ion transporter genes and possibly regulating the membrane potential (Fu et al. 2012). Hence, the *CsaRCI2* genes may also play important roles in the stabilization of plasma membrane potential, and then the improvement of ion homeostasis and membrane integrity under various abiotic stresses. Under abiotic stresses, plants can generate stress-related hormones such as ABA, which plays a key role in the adaptation of plants to various abiotic stress conditions through up-regulating the transcription of many stress-related genes (Fu et al. 2012; Vishwakarma et al. 2017). In this study, all the *CsaRCI2* genes were differentially regulated in response to ABA treatment, especially *CsaRCI2A* and *CsaRCI2D* (Fig. 7d), which might be due to the ABA-responsiveness *cis*-elements (ABRE) present in their promoter regions (Fig. 5). These results indicate that the expression of *CsaRCI2* genes may be regulated by ABA-dependent signaling pathways. Recent studies have also revealed that the transcripts of *AITMP1* and *AITMP2* from *Aeluropus littoralis* were dramatically increased under ABA treatment, and overexpressing either of them could confer abiotic stress tolerance in transgenic tobacco plants (Ben Romdhane et al. 2017; Ben-Romdhane et al. 2018). Therefore, *CsaRCI2* genes may play vital roles in plant response to abiotic stresses by maintaining plasma

membrane potential through an ABA dependent signaling pathway.

In the present study, we identified 4 *RCI2* family genes (*CsaRCI2A-CsaRCI2D*) from cucumber by a genome-wide search, and analyzed their basic classifications, protein and gene architectures, *cis*-elements in the promoter regions, molecular evolution, and tissue expression patterns. In addition, all the *CsaRCI2* genes were found to be remarkably responsive to cold, salt, drought, and ABA treatments, indicating that they can be used as potential targets for genetic manipulation to develop stress-resistant plants. Our findings lay a solid foundation for future functional characterization of *RCI2* family genes in cucumber.

Acknowledgements This work was funded by the Key Project of Youth Science Foundation of Jiangxi Province (20171ACB21025), and the National Natural Science Foundation of China (31460522 and 31660578).

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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